



Summer 2023

Exam Date: 10.08.2023

Exam Time 60 min.

MEP221s: Fluid Mechanics and Turbomachinery
Midterm Examination

The Exam Consists of **three** Questions on **Three** Pages

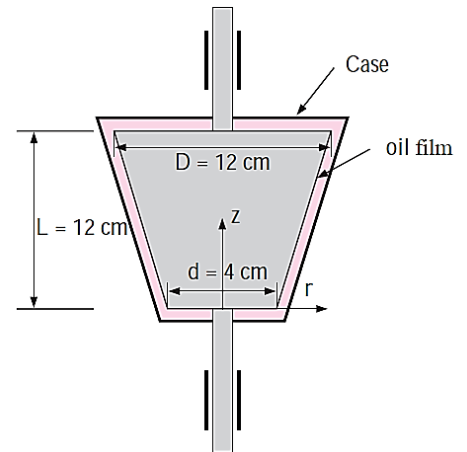
Total Marks: 25 Points

Student Name:

ID:

Question (1): (10 points)

A frustum-shaped body is rotating at a constant angular speed of 200 rad/s in a container filled with oil with $\mu = 0.1$ Pa.s. If the thickness of the oil film on all sides is 1.2 mm, determine the power required to maintain this motion.

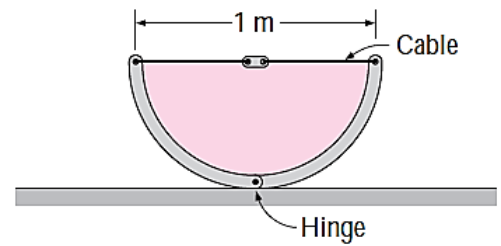


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Question (2): (8 points)

A water trough of a semicircular cross-section of radius 0.5 m consists of two symmetric parts hinged to each other at the bottom. The two parts are held together by a cable and turnbuckle placed every 3 m along the length of the trough. Calculate the tension in each cable when the trough is filled to the rim.

(Note: The centroid of a quarter-circle is $4R/3\pi$ away from the centerline)



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Question (3): (7 points)

For what uniform rotation speed in rpm about axis C will the mercury in the U-tube take the configuration shown?

